Architecture One-Pager — ITDX26

Observation-to-action pipeline; role boundaries

Layered architecture

Physical world: air · soil · mycelium · water · sound · light · movement · weather

Droid body layer: deterministic acquisition, calibration, local limits, health

Edge cortex: Jetson / NPU / TinyML — fingerprints, anomaly scores, NLM state, compression

Evidence contract: rooted observation — identity, time, location, calibration, quality, raw ref, hash

Mycorrhizae + DIRTNet: pub/sub, link selection, routing, store-forward, partition recovery

MINDEX: time series, graph, vector, raw refs, command records, Merkle epochs

NLM / FUSARIUM: state + pattern estimation, typed graph analysis

MYCA scenario runtime: role assignment, hypothesis generation, COA proposal

AVANI gate: deterministic admissibility — PASS / REVIEW / DENY / PAUSE

Bounded command intent → local droid safety controller → ACK + outcome evidence

Earth Simulator / CREP / replay: operator projection with uncertainty and labels

Role boundaries

- NLM estimates state and uncertainty. It does not act.
- FUSARIUM analyzes graphs and relationships. It does not act.
- MYCA orchestrates algorithms and proposes COAs. It does not directly actuate.
- AVANI constrains action; requires human approval for high-risk actions.
- MINDEX preserves lineage. It does not decide.
- CREP / Earth Simulator presents evidence. It does not conclude.

Product-compute rule

Compute runs on the droids and field gateway according to capability tier. The ITDX laptops are interfaces and evaluation surfaces. They are not the reason the product works and must not be described as the product architecture.

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